

LocalBrief

Private lecture & meeting notes
for Snapdragon-powered HP PCs

Record → transcribe locally → understand → act

Speech-to-text

Whisper via Qualcomm AI Hub
NPU-oriented Windows path

Note generation

Llama-v3.2-3B-Instruct
local structured output

Useful output

Summary • actions
quiz • glossary

The problem

People can record everything. They still cannot review everything.

Today

Record lecture / meeting

Send audio to a cloud tool

Wait for transcription

Copy useful parts into notes

The friction

Cloud upload may not be appropriate for every recording.

Too much output can be as unhelpful as too little.

A summary alone does not tell the user what to do next.

The opening

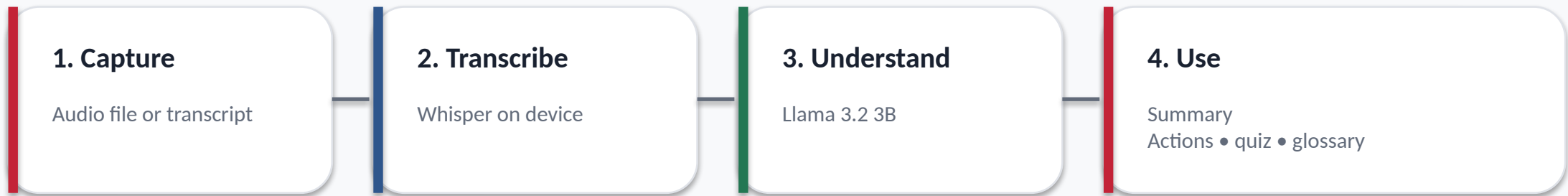
A laptop already has the compute needed for a local first pass.

Use AI where it helps: speech -> understanding
-> next actions.

The idea in one line

A small tool that turns spoken information into something you can study or act on.

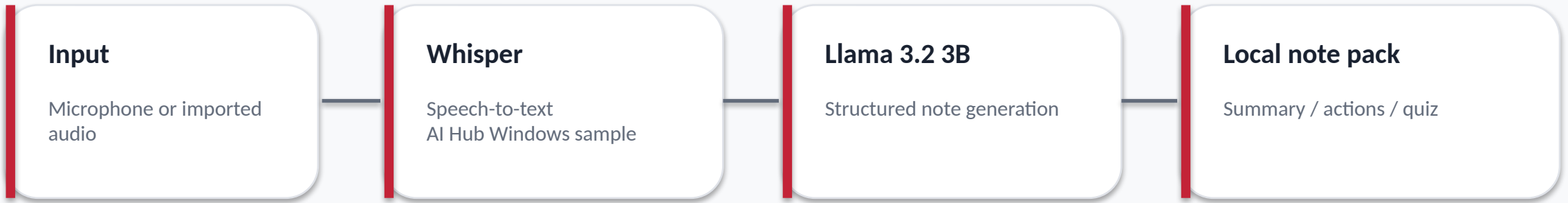
LocalBrief = a focused note-making pipeline, not a general chatbot.



The product stays simple on purpose: fewer screens, less noise, faster from recording to useful notes.

How it works

Two AI stages, one local workflow.



LOCAL-FIRST BOUNDARY

Audio and notes can stay on the PC.

The final build will show the actual runtime path and measurements on the target Snapdragon HP machine.

Why Snapdragon is part of the product

The hardware is used for the AI work, not added as a logo at the end.

NPU workload

Speech recognition

LLM inference

Local, low-latency AI stages

Measure the execution path on the final device

Qualcomm AI Hub

Pre-optimized model catalog

AI Hub Windows sample for Whisper

Llama-v3.2-3B-Instruct listed for Snapdragon X compute targets

Application result

No cloud API in the core note-generation path

One laptop handles capture, inference and review

Demo can show actual latency / tok/s

Qualcomm says Snapdragon X Series PCs support on-device AI through the Hexagon NPU; current and next-generation devices vary by model.

What the demo looks like

One short lecture. One clean note pack.

LocalBrief

Transcript

Today we are covering database indexing...
The goal is to reduce lookup time...
An index adds an extra structure...
For the assignment, finish the schema by Friday.

Note pack

Summary

Indexing reduces lookup work.

Action

Finish schema by Friday.

Quiz

What is an index?

Glossary

Index - lookup structure

The value is in the last step: turning a transcript into a usable artifact.

Implementation

A small codebase with a clear model/runtime boundary.

App layer

Streamlit UI
Input handling
Editable output
Export-ready note pack

AI layer

Whisper adapter
Llama adapter
Prompt + JSON output
Fallback demo mode

Runtime layer

ONNX Runtime GenAI
QNN / GenieX-compatible assets
Target Snapdragon NPU

Evidence layer

Environment check
Latency + tok/s benchmark
Actual device screenshots

The submission does not hard-code benchmark claims. It asks the target machine to prove the numbers.

What will be measured

Performance claims come from the target machine.

Transcription

Generation

Runtime path

Memory

Quality

Final benchmark panel

Audio sample __ sec

End-to-end __ sec

Generation __ tok/s

Memory __ MB / GB

Execution —

These blanks are intentional until the device run is complete.

Built around the four judging areas

The project is intentionally narrow enough to demo and measure well.

Technical implementation

Two-stage AI pipeline

Clear runtime boundary

Benchmark tooling

Targeted Snapdragon execution

Use case & innovation

Real note-taking problem

Local-first workflow

Summary + actions + study questions

Deployment

Small Python app

Browser UI

Demo mode + Snapdragon mode

No cloud API required for core flow

Presentation

Easy-to-follow demo

Architecture diagram

Measured results

Short explanation of the NPU path

The final pitch

LocalBrief is small on purpose: build it, run it, measure it, show it.

A practical local AI assistant for one of the most common things people do on a laptop: listen, learn and remember.

Problem

Too much spoken information; too little time to organize it.

Solution

Local speech + local language model -> useful notes.

Proof

A real Snapdragon HP demo with measured runtime results.

GitHub: <https://github.com/<your-username>/localbrief-snapdragon>

References: Qualcomm AI Hub Models, AI Hub Apps, Windows on Snapdragon AI documentation.